

1. TitanCore System Design Document (v2.4.1)

1.1 Executive Summary

TitanCore is a high-throughput, fault-tolerant microservices architecture designed to handle 50,000 TPS.

1.2 Architecture Overview

The system utilizes a hexagonal architecture. The core domain logic is isolated from external layers.

2.1 Component Specification: AuditLog (Shard 1)

2.1.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.1.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.1.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.1.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.2 Component Specification: PaymentGateway (Shard 2)

2.2.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the CQRS pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.2.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.2.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.2.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.3 Component Specification: RiskEngine (Shard 3)

2.3.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the SagaPattern pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.3.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.3.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.3.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.4 Component Specification: RiskEngine (Shard 4)

2.4.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.4.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.4.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.4.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.5 Component Specification: PaymentGateway (Shard 5)

2.5.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.5.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.5.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.5.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.6 Component Specification: RiskEngine (Shard 6)

2.6.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.6.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.6.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.6.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.7 Component Specification: AuditLog (Shard 7)

2.7.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the Sharding pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.7.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.7.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.7.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.8 Component Specification: AuditLog (Shard 8)

2.8.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.8.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.8.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.8.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.9 Component Specification: AuditLog (Shard 9)

2.9.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.9.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.9.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.9.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.10 Component Specification: AuditLog (Shard 10)

2.10.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the CircuitBreaker pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.10.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.10.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.10.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.11 Component Specification: PaymentGateway (Shard 11)

2.11.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.11.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.11.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.11.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.12 Component Specification: PaymentGateway (Shard 12)

2.12.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the EventSourcing pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.12.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.12.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.12.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.13 Component Specification: ReconModule (Shard 13)

2.13.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the SagaPattern pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.13.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.13.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.13.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.14 Component Specification: PaymentGateway (Shard 14)

2.14.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.14.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.14.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.14.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.15 Component Specification: ReconModule (Shard 15)

2.15.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.15.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.15.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.15.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.16 Component Specification: AuditLog (Shard 16)

2.16.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the CircuitBreaker pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.16.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.16.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.16.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.17 Component Specification: AuditLog (Shard 17)

2.17.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the Sharding pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.17.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.17.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.17.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.18 Component Specification: ReconModule (Shard 18)

2.18.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the SagaPattern pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.18.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.18.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.18.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.19 Component Specification: RiskEngine (Shard 19)

2.19.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.19.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.19.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.19.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.20 Component Specification: PaymentGateway (Shard 20)

2.20.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the CQRS pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.20.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.20.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.20.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.21 Component Specification: PaymentGateway (Shard 21)

2.21.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the SagaPattern pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.21.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.21.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.21.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.22 Component Specification: AuditLog (Shard 22)

2.22.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the CQRS pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.22.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.22.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.22.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.23 Component Specification: PaymentGateway (Shard 23)

2.23.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.23.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.23.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.23.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.24 Component Specification: PaymentGateway (Shard 24)

2.24.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.24.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.24.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.24.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.25 Component Specification: NotificationService (Shard 25)

2.25.1 Overview

The NotificationService module is responsible for handling high-volume traffic using the CQRS pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.25.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.25.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.25.4 Interface Definition

Endpoint: POST /api/v1/notificationservice/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for NotificationService continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.26 Component Specification: NotificationService (Shard 26)

2.26.1 Overview

The NotificationService module is responsible for handling high-volume traffic using the CircuitBreaker pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.26.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.26.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.26.4 Interface Definition

Endpoint: POST /api/v1/notificationservice/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for NotificationService continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.27 Component Specification: AuditLog (Shard 27)

2.27.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the CQRS pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.27.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.27.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.27.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.28 Component Specification: ReconModule (Shard 28)

2.28.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.28.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.28.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.28.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.29 Component Specification: NotificationService (Shard 29)

2.29.1 Overview

The NotificationService module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.29.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.29.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.29.4 Interface Definition

Endpoint: POST /api/v1/notificationservice/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for NotificationService continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.30 Component Specification: RiskEngine (Shard 30)

2.30.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the EventSourcing pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.30.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.30.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.30.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.31 Component Specification: RiskEngine (Shard 31)

2.31.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.31.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.31.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.31.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.32 Component Specification: ReconModule (Shard 32)

2.32.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the EventSourcing pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.32.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.32.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.32.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.33 Component Specification: NotificationService (Shard 33)

2.33.1 Overview

The NotificationService module is responsible for handling high-volume traffic using the SagaPattern pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.33.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.33.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.33.4 Interface Definition

Endpoint: POST /api/v1/notificationservice/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for NotificationService continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.34 Component Specification: AuditLog (Shard 34)

2.34.1 Overview

The AuditLog module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.34.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.34.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.34.4 Interface Definition

Endpoint: POST /api/v1/auditlog/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for AuditLog continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.35 Component Specification: RiskEngine (Shard 35)

2.35.1 Overview

The RiskEngine module is responsible for handling high-volume traffic using the CQRS pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.35.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.35.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

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Max Replicas: 50

2.35.4 Interface Definition

Endpoint: POST /api/v1/riskengine/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for RiskEngine continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.36 Component Specification: PaymentGateway (Shard 36)

2.36.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.36.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.36.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.36.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.37 Component Specification: LedgerService (Shard 37)

2.37.1 Overview

The LedgerService module is responsible for handling high-volume traffic using the CQRS pattern.

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It connects to the underlying Cassandra cluster on port 9042.

2.37.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.37.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.37.4 Interface Definition

Endpoint: POST /api/v1/ledgerservice/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for LedgerService continued...]

The internal buffer size is set to 1024KB. Memory management is handled via G1GC.

2.38 Component Specification: ReconModule (Shard 38)

2.38.1 Overview

The ReconModule module is responsible for handling high-volume traffic using the Sharding pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.38.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.38.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.38.4 Interface Definition

Endpoint: POST /api/v1/reconmodule/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for ReconModule continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.

2.39 Component Specification: PaymentGateway (Shard 39)

2.39.1 Overview

The PaymentGateway module is responsible for handling high-volume traffic using the CQRS pattern.

It connects to the underlying Cassandra cluster on port 9042.

2.39.2 Data Consistency

To ensure ACID properties, we utilize a two-phase commit where applicable, falling back to eventual consistency for read models.

2.39.3 Scaling Policy

Autoscaling is triggered when CPU > 75% or Memory > 80%.

Min Replicas: 3

Max Replicas: 50

2.39.4 Interface Definition

Endpoint: POST /api/v1/paymentgateway/process

Payload: JSON

Timeout: 5000ms

Retry Strategy: Exponential Backoff (Multiplier 1.5)

[Detailed Architecture notes for PaymentGateway continued...]

The internal buffer size is set to 1024KB. Memory management is handled via

G1GC.